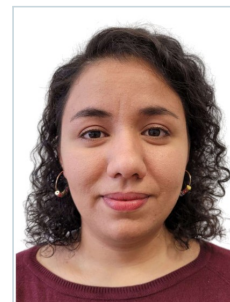


DR.-ING. SARAH OUERGHEMMI

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Profile

RF and embedded hardware engineer (Dr.-Ing., highest distinction) with 5+ years developing ultra-low-power radio front ends at 433 and 868 MHz, from circuit concept to validated prototype. Works in interdisciplinary teams on publicly funded consortium projects (BMBF/VDI, DAAD), supervising engineering students and contributing to successful funding proposals. Now looking for an industrial R&D or project leadership role.

Core competencies

RF front-end development	Ultra-low-power LNAs and matching networks (433/868 MHz), wideband and microstrip matching, noise figure, sensitivity and power budget optimization, ADS
Full hardware cycle	Schematic capture and PCB layout (Altium Designer, Fusion 360), prototype build, bench characterization (VNA, spectrum analyzer), MSP430/STM32 firmware
Projects and teams	Supervision of engineering students, project planning and reviews, laboratory and component management, DAAD funding proposals, multi-partner consortium projects (BMBF/VDI, DAAD) with international partners

Professional experience

03/2021 – present	Research Engineer — Wake-Up Receiver Development HTWK Leipzig (University of Applied Sciences), Leipzig, Germany • Development and validation of ultra-low-power RF front ends for wireless sensor networks within a research team, from circuit concept to bench characterization • Contributed a wake-up receiver front end operating below 10 μW while meeting sensitivity targets, enabling continuous connectivity in battery-powered sensor nodes • Full hardware cycle: 868 MHz LNA and matching network design, ADS simulation, PCB layout in Altium Designer / Fusion 360, prototype build and MSP430-based measurement • Supervision of final-year engineering students: task planning, weekly project reviews, laboratory resource and component management • Co-preparation of DAAD project proposals (Ta'Ziz, SDG initiatives) and representation of the institution at international conferences
2022 – 2026	RF Front-End Development — FUBE Project (BMBF/VDI funded) HTWK Leipzig, Leipzig, Germany • Responsible for the wake-up receiver RF front end within a multi-partner consortium developing lightweight radio sensors for space-based animal tracking • Ultra-low-power LNA and matching network design for deep-signal detection; sensitivity optimization for long-range space-to-sensor links • Provided the measurement basis used by the team for Doppler-based single-satellite localization
2022 – 2025	Partnership and Outreach — DAAD SDG Project CONCENTRATE, HTWK Leipzig & ENET'COM Sfax — co-organization of events and outreach between two partner institutions; project promotion at international conferences and support of career centre workshops.

Earlier experience

09 – 12/2020	Graduation Project, HTWK Leipzig, Germany — 868 MHz LNA architecture for wake-up receivers (ADS, Altium Designer).
2018 – 2019	Internships and student projects, Sagemcom Software & Technologies and PIVA Software, Tunisia — gateway test automation (Azure, ACS, TR-069, Jenkins), computer-vision recognition prototype (Python, OpenCV, TensorFlow), electric-meter production (SMD/THD assembly, test).

Results and project achievements

- Wake-up receiver front end brought below 10 μ W power consumption while meeting the sensitivity budget — the enabling condition for continuous connectivity in battery-powered nodes.
- 6 peer-reviewed publications on ultra-low-power LNA and wake-up receiver design (Electronics/MDPI, IEEE SENSORS, IEEE SSD); 43 citations, h-index 3.
- Several final-year engineering students supervised through to successful completion.
- Co-prepared DAAD proposals (Ta'Ziz, SDG) that secured funding for international project work.

Education

2021 – 2025	Doctoral Degree (Dr.-Ing.) — Embedded & Communication Systems, HTWK Leipzig, Germany & ENET'COM, Sfax, Tunisia — joint co-supervision on low-power RF front ends for wireless sensor networks; completed with the highest distinction and jury congratulations.
2017 – 2020	Engineering Degree in Electronics & Communication, ENET'COM, Sfax, Tunisia — embedded systems, communication networks, signal processing; ranked 2/60.
2014 – 2017	Preparatory Cycle Degree (Chemistry & Physics), Preparatory Institute for Engineering Studies of El Manar, Tunis, Tunisia.

Technical skills and tools

RF design	ADS (Advanced Design System); LNA and matching network design, noise figure, sensitivity and power budget analysis
PCB / schematic	Altium Designer, Fusion 360 Electronics; mixed-signal layout
Laboratory	VNA, spectrum analyzer, oscilloscope, logic analyzer, JTAG/SWD debugging
Embedded	MSP430, STM32 (ARM Cortex-M), DSP, Raspberry Pi, FPGA; bare-metal firmware, low-power design, peripheral drivers (HAL); UART, SPI, I ² C, CAN, TCP/IP, WSN
Programming / methods	C (embedded), C++, Python, Assembly, MATLAB, VHDL/VHDL-AMS; Git/GitHub, Agile, Scrum, UML, Linux

Languages

Arabic	Native speaker
English	Fluent (C1)
French	Fluent (C1)
German	Intermediate (B1)

Certifications, awards and engagement

Certifications / awards	LFS207 — Linux System Administration Essentials (Linux Foundation) with LIFT Training Scholarship, Women in Open Source; DAAD Scholarship (2020); IBM Artificial Intelligence Analyst — Mastery Award (2019)
Engagement	Handball team leader, ENET'COM (2018–2020); Communication Manager, Competitive Programming Club; organizing team, ACM Tunisian Collegiate Programming Contest (2019)